

THE ZUARI HIVE

Innovation Ecosystem Platform

Product Requirements Document

Prepared for Zuari Group

Simon India Limited · Zuari Industries Limited · Zuari Finserv Limited
Version 1.3 | Web Platform | Gemini AI Integration

Document Version	1.3 — Multi-User Role Handling (Option A)
Platform	Web Browser (Responsive)
AI Integration	Google Gemini API
Target Subsidiaries	Simon India Limited, Zuari Industries Limited, Zuari Finserv Limited
Document Status	Ready for Development

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1. Introduction

1.1 Purpose

This document defines the complete product requirements for The Zuari Hive — a multi-tenant, web-based innovation management platform serving three Zuari Group subsidiaries. It is intended as the primary reference for design, development, testing, and stakeholder alignment.

1.2 Product Vision

The Zuari Hive transforms how Zuari Group captures, evaluates, and implements employee ideas. By combining a frictionless submission experience with AI assistance, gamification, and a structured administrative workflow, the platform creates a culture of innovation — making it easy to contribute an idea and transparent to track its journey from 'spark' to corporate project.

1.3 Scope

Version 1.0 covers the following scope:

- Multi-tenant web application serving three subsidiary companies
- Three distinct user roles: Employee, Tenant Admin (Subsidiary Admin), and Superadmin (Central Team)
- Employee-facing 3-tab experience: The Spark, My Journey, and The Arena
- Administrative workflow: Global Triage, Delegation, Project Management
- AI-powered features: Idea Scribe (submission), AI Project Architect (planning), Template Builder
- Gamification engine: HIVE XP, levels, leaderboards
- Google Gemini API integration for all AI features

1.4 Out of Scope (v1.0)

- Native mobile applications (iOS / Android) — future phase
- Direct file uploads — users provide external links (OneDrive, SharePoint, Google Drive)
- Integration with HRMS, ERP, or payroll systems
- Real-time chat or internal messaging between users

2. User Roles & Permissions

2.1 Role Overview

The platform defines three user roles with distinct capabilities and views. A user belongs to exactly one subsidiary and one role.

Role	Description	Key Permissions
Employee	All staff members of any subsidiary. The primary idea contributors.	Submit ideas, track own ideas, view Arena, earn XP, upvote ideas
Tenant Admin	Subsidiary-level managers. One or more per company.	View all ideas in own subsidiary, initialize and manage assigned projects, manage subsidiary users
Superadmin	Central Zuari Group team. Platform-wide oversight.	View all ideas group-wide, triage and assign ideas to Tenant Admins, build form templates, view group analytics

2.2 Detailed Permission Matrix

Action	Employee	Tenant Admin	Superadmin
Submit idea	✓	✓	✓
View own ideas & status	✓	✓	✓
View all ideas (own subsidiary — assigned only)	✗	✓	✓
View all ideas (all subsidiaries)	✗	✗	✓
Upvote ideas in Arena	✓	✓	✗
Triage and assign ideas	✗	✗	✓
Initialize idea as project	✗	Assigned only	✗
Manage project tasks & timeline	✗	✓ (own)	✓ (all)
Build/edit form templates	✗	✗	✓
Manage subsidiary users	✗	✓ (own)	✓ (all)
View group-wide analytics	✗	Subsidiary only	✓
Award / adjust XP	✗	✗	✓

3. Multi-Tenant Architecture

3.1 Tenant Structure

The platform serves three tenants under one group umbrella. Each tenant corresponds to a subsidiary company:

- Simon India Limited
- Zuari Industries Limited
- Zuari Finserv Limited

3.2 Data Isolation Rules

The following data isolation rules must be enforced at the API and database level:

- Each idea, project, task, and user record is tagged with a `tenant_id` at creation
- Employees and Tenant Admins can only query and view records belonging to their own tenant
- Superadmins can query across all tenants with explicit `tenant_id` filters available
- Leaderboard data for 'Subsidiary Battle' aggregates XP per tenant (read-only cross-tenant access)
- Arena 'Discover Sparks' feed may show ideas from all tenants but hides sensitive internal detail fields
- Authentication must enforce tenant context — a user cannot impersonate or switch tenants

3.3 Branding & Templates

On login, the app detects the user's tenant and applies the correct:

- Primary brand color and logo (configurable per tenant by Superadmin)
- Default template catalog (Superadmin can assign templates to specific tenants or make them global)
- Email notification sender name (e.g., 'Simon India Innovation Hub')

4. Employee Experience — The 3-Tab System

The entire employee-facing UI is contained within three tabs. The navigation bar is always visible at the top. No other navigation surfaces are shown to Employees.

4.1 Tab 1 — The Spark (Idea Submission)

4.1.1 Landing State

When an Employee arrives at The Spark tab, they see two components:

- AI Scribe Input (Global Mode) — a large, prominent text area with placeholder copy: 'Describe your idea in plain language — our AI will find the right template for you.'

- Template Catalog — a full-width table below the AI input listing all available templates

4.1.2 Template Catalog

The template catalog is displayed as a professional table with the following columns:

- Template Name
- Category / Domain
- Best For (one-line description)
- No. of Questions
- Action button: 'Use This Template'

The catalog must include at minimum 20 templates across the following domains:

#	Template Name	Domain / Best For
1	Process Improvement	Engineering — Optimizing existing operational workflows
2	Cost Reduction Initiative	Finance — Identifying cost-saving opportunities
3	New Product Concept	Product — Proposing a new product or service line
4	Digital Transformation	IT — Digitizing manual or legacy processes
5	ESG / Sustainability	Environment — Green initiatives and sustainability goals
6	Customer Experience	CX — Improving customer journeys or NPS
7	Fintech Innovation	Finserve — New financial products or payment solutions
8	Safety & Compliance	HSE — Workplace safety improvements
9	Supply Chain Optimization	Logistics — Supplier or distribution improvements
10	Data & Analytics	BI — Leveraging data for better decisions
11	HR & People Culture	HR — Employee engagement and talent initiatives
12	Market Expansion	Strategy — Entering new markets or geographies
13	Automation & RPA	IT/Ops — Robotic process automation opportunities
14	Partnership Proposal	BD — External collaboration or JV suggestions
15	Revenue Growth	Sales — New revenue streams or pricing strategies

16	R&D / Experimental	Engineering — Research ideas without defined ROI
17	Regulatory & Policy	Legal — Responding to regulatory changes
18	Waste Reduction (Lean)	Manufacturing — Lean / Six Sigma improvements
19	Technology Adoption	IT — Evaluating and adopting new tech tools
20	Internal Knowledge Sharing	L&D — Knowledge management or training proposals

4.1.3 AI Scribe — Global Mode

Global Mode triggers when the user types in the top-level AI Scribe text area and clicks 'Analyze My Idea'.

AI Scribe Flow (Global Mode)

1. User enters free-text description in the global AI input field

2. System sends the text to Gemini API with a classification prompt

3. Gemini returns: (a) matched template ID, (b) pre-filled answers for each field

4. System opens the matched template form pre-populated with Gemini's suggestions

5. User reviews, edits fields as needed, and submits the idea

Fallback: If Gemini cannot match with confidence $\geq 70\%$, display top 3 template suggestions and let the user choose manually.

4.1.4 Template Forms

Each template form must contain a minimum of 7 questions. Questions follow a consistent structure:

- Question Title — clear, professional label
- Input Type — text field, text area, dropdown, multi-select, number field, URL field, or date picker
- Required / Optional flag
- Helper text — one-line guidance for the user
- AI Refine button (Local Mode) — per-field AI assistance

Example: Process Improvement Template questions:

- Problem Statement — What specific problem or inefficiency are you addressing? (Text area, Required)
- Current Process Description — Describe the current process step-by-step (Text area, Required)
- Proposed Solution — How would you change or improve this process? (Text area, Required)

- Expected Efficiency Gain — Estimated time/cost savings (% , hours/week, INR) (Text field, Required)
- Affected Departments — Which teams will be impacted? (Multi-select, Required)
- Implementation Complexity — Estimated difficulty: Low / Medium / High (Dropdown, Required)
- Resource Requirements — Tools, people, or budget needed (Text area, Optional)
- Supporting Documentation — Link to OneDrive, SharePoint, or Google Drive (URL field, Optional)
- Timeline Estimate — Expected duration to implement (Dropdown: <1 month, 1–3 months, 3–6 months, 6+ months, Required)

***i** All templates follow this 7–10 question structure. The Superadmin Architect Hub (Section 6.3) allows new templates to be generated via AI prompts.*

4.1.5 AI Scribe — Local Mode

Within any open template form, each text field and text area has an 'AI Refine' button. When clicked:

- System sends the current field value + template context to Gemini
- Gemini returns a refined, more professional or detailed version of the user's input
- A diff view shows original vs. suggested — user can Accept, Edit, or Dismiss
- Local Mode does not affect other fields

4.1.6 Link-Only File Attachment

No direct file upload is supported. All supporting documentation is attached as external links. The form provides a URL field that validates the link format and optionally checks accessibility. Accepted platforms: Microsoft OneDrive, SharePoint, and Google Drive.

4.1.7 Submission Confirmation

On successful submission:

- A success toast and confirmation screen are shown: 'Your Spark has been submitted! +10 HIVE XP earned.'
- The idea is assigned a unique Idea ID (e.g., ZH-2024-0042)
- The idea is routed to the Global Triage queue for Superadmin review
- The user's XP balance is updated immediately

4.2 Tab 2 — My Journey (Personal Roadmap)

4.2.1 Overview

My Journey is the employee's personal dashboard showing all ideas they have ever submitted, along with live status and progress tracking.

4.2.2 Idea Status States

Status	Description	Who Sets This
Submitted	Idea received, awaiting Superadmin review	System (automatic)
Under Screening	Superadmin is reviewing the idea	Superadmin
Needs Clarification	Superadmin has requested more information from submitter	Superadmin
Assigned	Idea delegated to a Tenant Admin for project execution	Superadmin
In Development	Tenant Admin has initialized the project; milestones active	Tenant Admin
Completed	Project successfully delivered	Tenant Admin / Superadmin
Closed — Not Feasible	Idea rejected with reason provided	Superadmin
Closed — Duplicate	Similar idea already exists; submitter notified	Superadmin
Closed — Out of Scope	Idea falls outside group innovation mandate	Superadmin

4.2.3 Idea Card View

Each idea in My Journey is displayed as a card containing:

- Idea ID and title
- Template used
- Submission date
- Current status badge (color-coded)
- Progress bar (visible once idea reaches 'In Development' status)
- Milestone checklist — live list of tasks added by the Admin, showing completed/pending state
- XP earned from this idea (cumulative)

i The progress bar calculates completion as: $(\text{completed milestones} / \text{total milestones}) \times 100$. Employees see milestone names but not internal admin notes. If an idea is closed while In Development (e.g. cancelled), the progress bar freezes at its last value and a 'Closed' banner overlays the card. No further XP is awarded after closure.

4.3 Tab 3 — The Arena (Gamification & Social)

4.3.1 Overview

The Arena is the community and competition hub. It contains three main sub-sections accessible via sub-navigation tabs within Tab 3.

4.3.2 Subsidiary Battle Leaderboard

A ranked table showing each of the three subsidiaries by total aggregate XP earned by all their employees. Columns: Rank, Company Name, Total HIVE XP, Total Ideas Submitted, Ideas In Development, Top Contributor Name.

4.3.3 Individual Leaderboard

A ranked table of all employees group-wide (cross-tenant). Columns: Rank, Employee Name, Company, Level, Total XP, Ideas Submitted, Ideas Approved. The current logged-in user's row is highlighted. Filterable by subsidiary.

4.3.4 Discover Sparks (Innovation Feed)

A paginated, card-based feed of publicly visible ideas submitted by other employees. Only ideas that have passed Superadmin screening (status: Assigned, In Development, or Completed) are shown — pre-triage or rejected ideas are never surfaced here.

Feed controls:

- Search bar — keyword search across idea title and summary
- Filter by Company — All, Simon India, Zuari Industries, Zuari Finserv
- Filter by Category — dropdown matching template domain tags
- Sort by — Most Recent (default), Most Upvoted, Most Active (recently updated)

Each card shows:

- Idea title and one-line summary
- Submitting employee's name and level badge
- Company subsidiary
- Template category tag
- Upvote count
- Upvote button

4.3.5 Upvote & Similar Problem Logic

Employees and Tenant Admins can upvote ideas in the Discover Sparks feed. Superadmins cannot upvote (their role is triage, not community voting). When a user clicks Upvote, a modal appears:

Upvote Modal Logic

Modal Title: Thanks for your support!

Modal Body: Why are you upvoting this idea?

- Option A — 'I like this idea' → Records upvote server-side, awards +2 XP to upvoter, closes modal, updates upvote count in UI
- **Option B — 'I have a similar problem' → Records upvote server-side and awards XP FIRST, then redirects user to The Spark tab with the same template pre-loaded. The upvote API call must complete (or fire-and-forget with confirmation) before navigation occurs to prevent XP loss.**

Anti-gaming rule: A user may only upvote any single idea once. The upvote button is disabled and shows a checkmark after voting. A user earns a maximum of 10 upvote XP events per day (+2 XP × 10 = +20 XP/day cap from upvoting).

5. Gamification — HIVE XP System

The HIVE XP system is designed to reward genuine innovation behaviour — submitting quality ideas, seeing them through to completion, and engaging constructively with the community. XP is cumulative and never decreases. Levels are permanent once reached.

5.1 XP Award Rules

Trigger Event	XP Awarded	Awarded To	Notes / Limits
Idea submitted (any template)	+10 XP	Submitter	Once per idea submission
Idea passes Superadmin screening (approved)	+20 XP	Submitter	Once per idea
Idea assigned to Tenant Admin (delegated)	+10 XP	Submitter	Once per idea
Project milestone completed by Admin	+15 XP	Submitter	Max 5 milestone XP events per project (75 XP cap). Admin must set milestones before project starts — cannot add retroactively.
Project marked as Completed	+50 XP	Submitter	Once per project
Idea receives 5 upvotes	+5 XP	Submitter	One-time threshold bonus

Idea receives 20 upvotes	+10 XP	Submitter	One-time threshold bonus; stacks with 5-upvote bonus
Upvoting another idea	+2 XP	Upvoter	Max 10 upvote XP events per user per day (20 XP/day cap). One upvote per idea per user — cannot upvote own ideas.

i Maximum XP earnable from a single idea lifecycle: 10 (submit) + 20 (approved) + 10 (assigned) + 75 (milestones, capped) + 50 (completed) + 15 (upvote bonuses) = 180 XP. This cap prevents any single idea from distorting the leaderboard.

5.2 User Levels & Progression

Levels are calculated in real-time from the user's cumulative XP total. Level-up events trigger an in-app congratulatory toast notification and update the user's badge immediately.

Level	Title	XP Required	Badge Colour	Unlocks & Benefits
1	Explorer	0 XP	Grey	Full submission access. Level badge shown on leaderboard and idea cards.
2	Innovator	100 XP	Blue	XP progress bar becomes visible in profile. Eligible to appear on Individual Leaderboard.
3	Catalyst	300 XP	Gold	Featured 'Catalyst' badge displayed on all Discover Sparks cards. Ideas surfaced higher in default feed sort.
4	Visionary	750 XP	Purple	Ideas flagged with a Visionary star in Superadmin Triage queue as a soft signal. Exclusive Visionary badge on leaderboard.

5.3 XP & Level Display — Where Badges Appear

The following table defines exactly where level badges and XP information are surfaced in the UI, so the developer has a complete rendering spec:

UI Location	What Is Shown
User profile / account menu (top-right header)	Level badge icon + title (e.g. 'Catalyst'), total XP, XP progress bar to next level showing X / Y XP
Individual Leaderboard row	Level badge colour-dot + title next to the employee's name
Subsidiary Leaderboard	No individual badges — aggregate XP total and top contributor name only
Discover Sparks feed — idea card	Submitter's level badge and title shown beneath their name
My Journey — idea card	XP earned from this specific idea shown as a small tag (e.g. '+55 XP earned')
Level-up toast notification	Full-screen overlay: 'You are now a [Level Title]!' with badge animation, dismissed on tap/click
Triage Queue (Superadmin)	Visionary star icon next to submitter name for Level 4 users only

5.4 Subsidiary Battle — Calculation Rules

The Subsidiary Battle leaderboard competes on two dimensions simultaneously:

- All-Time XP Total — cumulative XP earned by all employees of that subsidiary since platform launch. This is the primary ranking metric.
- Monthly XP — XP earned by the subsidiary in the current calendar month. Shown as a secondary column to highlight recent momentum. Resets on the 1st of each month.
- The leaderboard refreshes every 15 minutes (server-side cache). It does not update in real-time to avoid rendering thrash on high-activity days.
- If two subsidiaries have identical All-Time XP (edge case), Monthly XP is the tiebreaker. If still tied, the subsidiary with more total ideas submitted ranks higher.

5.5 Anti-Gaming & Integrity Rules

- Milestone XP cap: Maximum 5 milestone XP award events per project (75 XP total from milestones). Admin must define all milestones before the project status moves to In Development — milestones cannot be added after the project starts to prevent inflation.
- Self-upvoting: Users cannot upvote their own ideas. The upvote button is hidden on a user's own idea cards.
- Daily upvote XP cap: +2 XP per upvote, maximum 10 qualifying upvotes per user per day (20 XP/day max from upvoting).
- Duplicate submissions: If a Superadmin closes an idea as 'Duplicate', no XP is awarded for the submission. The +10 XP is reversed if already awarded (XP deduction recorded as a negative XP transaction).
- XP is non-transferable between users and cannot be gifted, pooled, or manually adjusted by Tenant Admins. Only Superadmins can issue manual XP corrections, with a mandatory note recorded in the XP transaction log.

- All XP events are recorded in the XP Transaction table with timestamp, trigger_event, and reference_id for full auditability.
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6. Administrative Workflow

6.0 Multi-User Handling Within Roles

Both the Central Team (Superadmin) and each subsidiary team (Tenant Admin) may have multiple people working simultaneously under the same role tier. The platform handles this without introducing new roles, using two lightweight mechanisms:

Central Team (Multiple Superadmins)

All Superadmin users share equal access to the Global Triage Queue. To prevent two people from reviewing the same idea at the same time, each idea in the queue has a 'Claim' action.

When a Superadmin clicks 'Claim', their name is assigned as the Reviewing Officer for that idea. The idea row displays their name and is visually tagged as 'In Review by [Name]'. Other Superadmins can still open and read the idea but cannot take action on it while it is claimed.

A Superadmin can release their claim (returning the idea to unclaimed status) or reassign it to a colleague. Ideas that have been claimed for more than 5 business days with no action are automatically released back to the unclaimed queue and the original reviewer is notified.

Rule: Only the claiming Superadmin (or any Superadmin if unclaimed) can approve, delegate, request clarification, or close an idea.

Subsidiary Team (Multiple Tenant Admins / Project Workers)

A subsidiary may have multiple Tenant Admin accounts (e.g. a project manager, a developer lead, a business analyst). All Tenant Admins within the same subsidiary share visibility of all ideas assigned to that subsidiary.

When initializing a project, the Tenant Admin selects a Project Owner from the list of Tenant Admins in their subsidiary. The Project Owner is the primary point of contact for that project and receives all project-related notifications. Other Tenant Admins in the subsidiary can view the project but only the Project Owner can edit milestones, change project status, or complete tasks.

Individual tasks within a project can be assigned to any employee in the subsidiary (regardless of role) using the 'Assigned To' field on each task. This is how developers, business users, and other contributors are pulled into the work without needing an admin account.

Rule: Task completion (and therefore XP triggering) can only be toggled by the Project Owner or any Tenant Admin — not by the task assignee themselves.

i This approach keeps the 3-role structure clean while fully supporting teams of any size working under each role. Sub-roles, team hierarchies, or permission groups are explicitly out of scope for v1.0 and can be revisited in a future release based on actual usage patterns.

6.1 Superadmin — Global Triage

6.1.1 Triage Queue

All submitted ideas appear in the Global Triage queue. The queue is a sortable, filterable table with the following columns:

- Idea ID — unique identifier
- Title — idea summary
- Submitted By — employee name and company
- Template — template used
- Submitted On — date
- Upvotes — number from Arena
- Reviewing Officer — name of the Superadmin who has claimed this idea, or 'Unclaimed' if available
- Status — current triage state
- Actions — Claim (if unclaimed), Review (if claimed by self), Release, Assign, Close

6.1.2 Idea Review Panel

Clicking Review opens a full-panel view showing all form responses for the idea. The Superadmin can:

- Add an internal note (not visible to the submitter)
- Flag the idea as Feasible / Not Feasible / Needs Clarification
- Request clarification from the submitter (triggers an email and in-app notification)
- Assign to a Tenant Admin (triggers delegation flow)
- Close the idea with a reason (Not Feasible, Duplicate, Out of Scope)

i Clarification Request Flow: When the Superadmin requests clarification, the idea status changes to 'Needs Clarification'. The submitter receives a notification with the Superadmin's question. The submitter responds via a text field shown on their My Journey idea card. On submission of the response, the idea reverts to 'Under Screening' status and the Superadmin is notified. The original question and the submitter's response are both stored in the idea's internal notes log.

6.1.3 Delegation Flow

When a Superadmin assigns an idea to a Tenant Admin:

- The Superadmin selects the target subsidiary and a specific Tenant Admin from a dropdown
- An optional message can be included with the assignment
- The idea status changes to 'Assigned'
- The Tenant Admin receives an in-app notification and an email alert
- The Tenant Admin can now see the idea in their Project Portfolio and is able to initialize it as a project

i *Tenant Admins cannot take action on any idea until it has been formally assigned by the Superadmin. This is a hard system rule enforced at the API level.*

6.2 Tenant Admin — Project Portfolio

6.2.1 Portfolio Overview

The Tenant Admin's main view is a Project Portfolio dashboard. Tenant Admins can only see ideas that have been formally assigned to them by the Superadmin — they do not have visibility into unassigned or under-screening ideas within their subsidiary. This prevents premature exposure of triage decisions.

The Portfolio dashboard shows:

- All ideas assigned to them (status: Assigned — awaiting initialization)
- All active projects (status: In Development)
- Completed projects archive
- Summary metrics: Total projects, On Track, At Risk, Completed

6.2.2 Initializing a Project

To initialize an assigned idea as a project, the Tenant Admin clicks 'Initialize Project'. This opens a setup form:

- Project Name (pre-filled from idea title, editable)
- Project Description (pre-filled from idea summary, editable)
- Project Owner — dropdown of all Tenant Admin accounts in the subsidiary. Defaults to the user initializing the project. The Project Owner receives all project notifications and has edit rights.
- Project Start Date
- Estimated End Date
- Team Members — multi-select of all employees in the subsidiary. Selected members can be assigned to individual tasks.
- Priority Level — Critical / High / Medium / Low
- Use AI Project Architect button (optional)

6.2.3 AI Project Architect

If the Tenant Admin clicks 'Use AI Project Architect', the system sends the original idea's goals and context to Gemini with a project planning prompt. Gemini returns:

- A suggested multi-week implementation timeline
- A list of 5–10 recommended milestones with descriptions
- Suggested team roles for each phase

The Admin reviews the AI-generated plan and can accept all, accept individual milestones, or manually edit before saving.

6.2.4 Task & Milestone Management

Within each active project, the Admin manages a task board. Each task/milestone contains:

- Task Title
- Description
- Assigned To (team member from subsidiary)
- Start Date and Due Date
- Status — Not Started, In Progress, Completed, Blocked
- Completion toggle — marking complete triggers XP award to idea submitter and updates their progress bar

6.3 Superadmin — Architect Hub (Template Builder)

6.3.1 Purpose

The Architect Hub allows Superadmins to create new innovation form templates without writing code. Templates are stored globally and can be made available to all or specific subsidiaries.

6.3.2 AI Template Generation

The Superadmin enters a prompt describing the type of innovation they want to capture (e.g., 'Create a template for AI adoption proposals in our fertilizer manufacturing division'). Gemini returns:

- A suggested template name and category
- 7–10 technical and strategic questions with suggested input types
- Helper text for each question

The Superadmin can then edit any field, add or remove questions, assign the template to tenants, and publish it to the catalog.

6.3.3 Template Schema

Each template record stores:

- `template_id` (UUID)
- `name`, `category`, `description`, `best_for`

- tenant_visibility — 'global' or array of tenant IDs
- is_active (boolean)
- fields[] — array of field objects: { id, label, type, required, helper_text, options[] }
- created_by (Superadmin user ID), created_at, updated_at

7. Feature Requirements Summary

Features are prioritized as: P0 = Must Have (MVP), P1 = Should Have (v1.0), P2 = Nice to Have (post-launch)

ID	Feature	Description	Priority
F-01	User Authentication	Email/password login with tenant-aware session management	P0
F-02	Multi-Tenant Data Isolation	All records tagged with tenant_id; enforced at API level	P0
F-03	Template Catalog	Table of 20+ templates with domain, description, and question count	P0
F-04	Idea Submission Form	7-10 question form per template with required/optional fields	P0
F-05	AI Scribe — Global Mode	Free-text input → Gemini classifies → pre-fills correct template	P0
F-06	AI Scribe — Local Mode	Per-field AI refine button within any open template form	P1
F-07	Link Attachment	URL fields for OneDrive/SharePoint/Drive with format validation	P0
F-08	Submission XP Award	+10 XP on idea submission, instant balance update	P0
F-09	My Journey Dashboard	Personal roadmap with status cards for all submitted ideas	P0
F-10	Progress Bar & Milestones	Live progress bar + milestone checklist once idea is In Development	P0
F-11	Idea Status Engine	9 statuses with automated transitions, clarification response flow, and notifications	P0
F-12	Arena — Subsidiary Leaderboard	Company-level XP aggregate rankings table	P0
F-13	Arena — Individual Leaderboard	Cross-tenant individual rankings with subsidiary filter	P0
F-14	Discover Sparks Feed	Paginated public feed of ideas with upvote functionality	P1

F-15	Upvote Modal — Similar Problem	Modal with Like / Similar Problem routing to pre-loaded Spark	P1
F-16	HIVE XP Engine	Full XP award rule system across all trigger events	P0
F-17	Level Progression	4 levels (Explorer → Visionary) with threshold-based promotion	P1
F-18	Global Triage Queue	Superadmin view of all submitted ideas, sortable/filterable	P0
F-19	Idea Review Panel	Full-panel idea detail view with notes, flags, and actions	P0
F-20	Delegation Flow	Assign idea to Tenant Admin with message and notification	P0
F-21	Project Portfolio	Tenant Admin project overview with metrics summary	P0
F-22	Project Initialization	Form to create project from assigned idea	P0
F-23	AI Project Architect	Gemini-generated timeline, milestones, and team roles	P1
F-24	Task & Milestone Board	Task management with completion toggling and date tracking	P0
F-25	Architect Hub	Superadmin template builder with AI generation support	P1
F-26	Email Notifications	Alerts on: submission, assignment, status change, milestone	P1
F-27	In-App Notifications	Bell icon with unread count and notification drawer	P1
F-28	Tenant Branding Config	Per-tenant primary color and logo management	P2
F-29	Superadmin Analytics	Group-wide innovation metrics and trend charts	P2
F-30	Clarification Request & Response	Superadmin requests info; submitter responds inline on My Journey card	P1
F-31	XP Profile Widget	Header profile showing level badge, total XP, and progress bar to next level	P0
F-32	Level-Up Toast Overlay	Full-screen congratulatory animation on level-up event	P1
F-33	Discover Sparks Search & Filter	Keyword search + filter by company/category + sort controls on feed	P1
F-34	Upvote Anti-Gaming Controls	One upvote per user per idea; no self-upvoting; daily XP cap enforced at API	P0
F-35	Milestone Lock	Milestones locked from addition after project moves to In Development	P0

F-36	XP Audit Log	Full transaction history with trigger, reference, and admin notes for manual adjustments	P1
F-37	Session Management	JWT access (8hr) + refresh (7 days) tokens; silent refresh; auto-logout on expiry	P0
F-38	Triage Claim & Reviewer Assignment	Superadmins claim ideas to prevent simultaneous review; 5-day auto-release on inactivity	P0
F-39	Project Owner Assignment	Tenant Admin designates a Project Owner on initialization; owner has edit rights and receives all notifications	P0

8. AI Integration Specifications (Google Gemini)

8.1 General Principles

- Model: Google Gemini 1.5 Pro (or latest stable API version at time of development)
- All AI calls are server-side — the Gemini API key is never exposed to the browser
- Every AI response is suggestion-only — the user always has final control and can override
- AI calls must have a maximum wait time of 15 seconds; display a loading state beyond 3 seconds
- Graceful fallback: if Gemini returns an error or timeout, the user can proceed manually without AI
- Confidence threshold (applies to all AI classification features): if Gemini returns a confidence score below 70%, the system must not auto-apply the result. Instead, surface the top suggestions and ask the user to confirm or choose manually. For generative features (Local Refine, Project Architect, Template Builder), there is no confidence gate — output is always shown as a suggestion.
- All AI-generated content is clearly labelled with an 'AI Suggested' badge in the UI so users know what was human-authored vs. AI-assisted

8.2 AI Feature Specifications

Feature	Input to Gemini	Expected Output	User Action
AI Scribe Global	Free-text idea description + list of all template names & descriptions	JSON: { template_id, confidence_score, pre_filled_fields[] }	Review pre-filled form, edit, submit
AI Scribe Local	Field label + helper text + user's current input	Refined version of the field text	Accept / Edit / Dismiss diff view
AI Project Architect	Idea title + goals + submitted form answers	JSON: { timeline_weeks, milestones[], team_roles[] }	Accept all / edit individual / discard

Architect Hub Template Builder	Superadmin's natural language description of the template purpose	JSON: { template_name, category, fields[] with labels, types, helpers }	Edit fields, set visibility, publish
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9. Notifications

9.1 In-App Notifications

A bell icon in the top navigation bar shows an unread notification count badge (red dot with number). Clicking opens a right-side drawer with a list of timestamped notifications, most recent first. Individual notifications are marked read on click; a 'Mark all as read' button clears all. Notifications persist for 90 days then are auto-archived.

The following events generate in-app notifications:

- Idea submitted — confirmation to submitter: 'Your Spark [ZH-XXXX] has been received. +10 XP!'
- Idea status changed to Under Screening — to submitter: 'Your idea is being reviewed by the central team.'
- Clarification requested — to submitter: 'The central team has a question about [Idea Title]. Tap to respond.'
- Idea assigned to Tenant Admin — to submitter: 'Great news! Your idea has been assigned for development.' Also to Tenant Admin: 'A new idea has been assigned to you: [Idea Title].'
- Idea closed — to submitter: 'An update on your idea [ZH-XXXX]: [close reason].'
- Milestone completed — to submitter: 'Progress update: [Milestone Name] has been completed on your project. +15 XP!'
- Project completed — to submitter: 'Your idea has been fully implemented! +50 XP! 🎉'
- Level-up — to user: 'Congratulations! You are now a [Level Title]!' (triggers full-screen overlay, see Section 5.3)
- Upvote milestone reached (5 or 20 upvotes) — to submitter: 'Your idea just reached [X] upvotes! +[Y] XP'

9.2 Email Notifications

The following events trigger email notifications:

Trigger	Recipient	Subject Line
Idea submitted	Submitter (confirmation)	<i>Your Spark ZH-XXXX has been received!</i>
Idea assigned to Tenant Admin	Tenant Admin	<i>New idea assigned to you: [Idea Title]</i>

Idea status changed	Submitter	<i>Update on your idea: [Idea Title]</i>
Clarification requested	Submitter	<i>Your input is needed: [Idea Title]</i>
Milestone completed	Submitter	<i>Progress update on your project: [Idea Title]</i>
Project completed	Submitter	<i>Your idea has been implemented! 🎉</i>

10. Non-Functional Requirements

10.1 Performance

- Page load time: < 2 seconds on standard broadband connection
- API response time: < 500ms for non-AI endpoints
- AI Gemini response: display loading state after 3 seconds; hard timeout at 15 seconds
- Leaderboard data: refreshed every 15 minutes (can be cached)

10.2 Security

- All API endpoints require authenticated JWT tokens with tenant_id and role embedded in the payload
- Role-based access control enforced server-side — never rely on client-side role checks alone
- JWT access tokens expire after 8 hours; refresh tokens expire after 7 days. On access token expiry, the client silently refreshes using the refresh token. On refresh token expiry, the user is redirected to login.
- Gemini API key stored as server-side environment variable only — never returned to the client
- User passwords hashed using bcrypt (min cost factor 12)
- HTTPS enforced on all routes; HTTP redirects to HTTPS
- Rate limiting on AI endpoints: max 20 AI calls per user per hour
- Rate limiting on login endpoint: max 10 failed attempts per IP per 15 minutes, then temporary lockout
- All sensitive actions (idea close, XP manual adjust, user deactivation) are logged in an audit trail with user ID, timestamp, and action details
- Cross-tenant data access attempts (e.g. a user querying another tenant's idea_id) must return 403 Forbidden, not 404, to prevent enumeration attacks

10.3 Scalability

- Architecture should support horizontal scaling (stateless API server)
- Database should use indexed queries on `tenant_id`, `user_id`, `idea_id`, and `status` fields
- Image/asset storage should use a CDN-backed object storage solution

10.4 Accessibility

- WCAG 2.1 Level AA compliance for all user-facing pages
- All form fields must have associated labels
- Color is never the sole indicator of status — always pair with text or icon

10.5 Browser Support

- Chrome (latest 2 versions)
- Firefox (latest 2 versions)
- Safari (latest 2 versions)
- Edge (latest 2 versions)
- Mobile responsive layout for Chrome and Safari on iOS and Android

11. Core Data Models

i *These are logical models for developer reference. The tech stack choice determines the actual database implementation.*

11.1 Tenant

- `tenant_id` (UUID, PK)
- `name` — e.g. 'Simon India Limited'
- `slug` — e.g. 'simon-india' (used in subdomain or URL path routing)
- `primary_color` (hex string), `logo_url`
- `is_active` (boolean)
- `created_at`

11.2 User

- `user_id` (UUID, PK)
- `email` (unique), `password_hash`, `full_name`
- `tenant_id` (FK → Tenant)

- role — ENUM: employee | tenant_admin | superadmin
- xp_total (integer, default 0)
- level — ENUM: explorer | innovator | catalyst | visionary (computed from xp_total, can be stored as a denormalized cache)
- created_at, updated_at, last_login_at, is_active (boolean)

11.3 Idea

- idea_id (UUID, PK), idea_ref (e.g. ZH-2024-0042, auto-generated sequential)
- title, summary (short description max 160 chars)
- template_id (FK → Template)
- submitted_by (FK → User)
- tenant_id (FK → Tenant)
- form_responses (JSON — field_id: value pairs)
- status — ENUM: submitted | screening | needs_clarification | assigned | in_development | completed | closed
- close_reason (nullable string)
- assigned_to (FK → User — Tenant Admin, nullable)
- reviewing_officer_id (FK → User — Superadmin who has claimed this idea for triage, nullable)
- claimed_at (timestamp, nullable — used to enforce the 5 business day auto-release rule)
- upvote_count (integer, denormalized for performance)
- visibility — ENUM: public | private (default: public after screening)
- superadmin_notes (text, internal only)
- created_at, updated_at

11.4 Project

- project_id (UUID, PK)
- idea_id (FK → Idea, unique — one project per idea)
- tenant_id (FK → Tenant)
- initialized_by (FK → User — Tenant Admin who created the project)
- project_owner_id (FK → User — Tenant Admin responsible for this project; may differ from initialized_by)
- team_member_ids (array of User UUIDs — employees added to the project; used to populate task assignment dropdowns)
- project_name, description, priority — ENUM: critical | high | medium | low
- start_date, estimated_end_date, actual_end_date (nullable)
- status — ENUM: active | completed | on_hold | cancelled
- milestones_locked (boolean, default false) — set to true when project moves to In Development; prevents new milestone additions
- created_at, updated_at

11.5 Milestone / Task

- task_id (UUID, PK)
- project_id (FK → Project)
- title, description
- assigned_to (FK → User, nullable)
- start_date, due_date, completed_at (nullable)
- status — ENUM: not_started | in_progress | completed | blocked
- xp_trigger_fired (boolean, default false) — prevents double-awarding XP if task is reopened and re-completed
- sort_order (integer) — for drag-and-drop reordering on the task board

11.6 Upvote

- upvote_id (UUID, PK)
- idea_id (FK → Idea)
- user_id (FK → User)
- upvote_type — ENUM: like | similar_problem
- created_at
- UNIQUE constraint on (idea_id, user_id) — enforces one upvote per user per idea at the database level

11.7 XP Transaction

- xp_id (UUID, PK)
- user_id (FK → User)
- amount (integer — positive for award, negative for reversal)
- trigger_event — ENUM: idea_submitted | idea_approved | idea_assigned | milestone_completed | project_completed | upvote_received_5 | upvote_received_20 | upvote_given | manual_adjustment | duplicate_reversal
- reference_id (UUID — idea_id, task_id, or upvote_id depending on trigger)
- note (nullable string — required for manual_adjustment events)
- created_by (FK → User — system or Superadmin for manual adjustments)
- created_at

11.8 Notification

- notification_id (UUID, PK)
- user_id (FK → User — recipient)

- type — ENUM: idea_received | status_change | clarification_request | level_up | milestone_done | project_complete | upvote_milestone | assignment
- title, body (short text for drawer display)
- reference_id (UUID — idea_id or project_id)
- is_read (boolean, default false)
- created_at, expires_at (90 days from created_at)

12. Open Questions & Decisions for Developer

#	Question	Notes / Options
1	What is the SSO / login strategy? Email+password only, or SSO via Microsoft/Google?	<i>Zuari likely uses Microsoft 365 — MSAL/Azure AD SSO may be preferred</i>
2	Is the Arena feed fully public across subsidiaries, or opt-in per idea?	<i>Submitter may want to keep idea private before triage</i>
3	Should the Discover Sparks feed show ideas pre-triage or only after screening?	<i>Recommend: only show approved/assigned ideas to avoid surfacing rejected content</i>
4	What email provider will be used for transactional emails?	<i>Options: SendGrid, AWS SES, Mailgun, or SMTP relay</i>
5	Should XP be retroactively awarded if rules change?	<i>Recommend: no retroactive changes — document version policy</i>
6	Are there any DPDPA (India data protection) or internal data classification requirements?	<i>Especially relevant for employee names appearing on cross-tenant leaderboards</i>
7	Will Tenant Admins be able to invite new employees, or is user provisioning centralized?	<i>Recommend: Superadmin creates accounts; Tenant Admin can deactivate</i>
8	What is the expected user volume per subsidiary at launch?	<i>Helps size infrastructure and set rate limits appropriately</i>

13. Glossary

Spark	An employee-submitted innovation idea
Hive XP	The platform's gamification points currency
Triage	The Superadmin process of reviewing and routing incoming ideas
Delegation	The act of a Superadmin assigning an idea to a Tenant Admin for project execution

Arena	The social and gamification tab where employees compete and explore ideas
Tenant	A subsidiary company (Simon India, Zuari Industries, or Zuari Finserv)
Architect Hub	The Superadmin tool for building new template schemas using AI
AI Scribe	The AI-powered input assistant that classifies and pre-fills idea forms
AI Project Architect	The AI tool that generates project plans from idea goals
Milestone	A key deliverable checkpoint within a project, tracked by the Tenant Admin
Visionary	The highest user level (750+ XP), indicated by a special badge in the Arena
Discover Sparks	The public innovation feed in the Arena where employees browse and upvote ideas

END OF DOCUMENT

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